Recursion

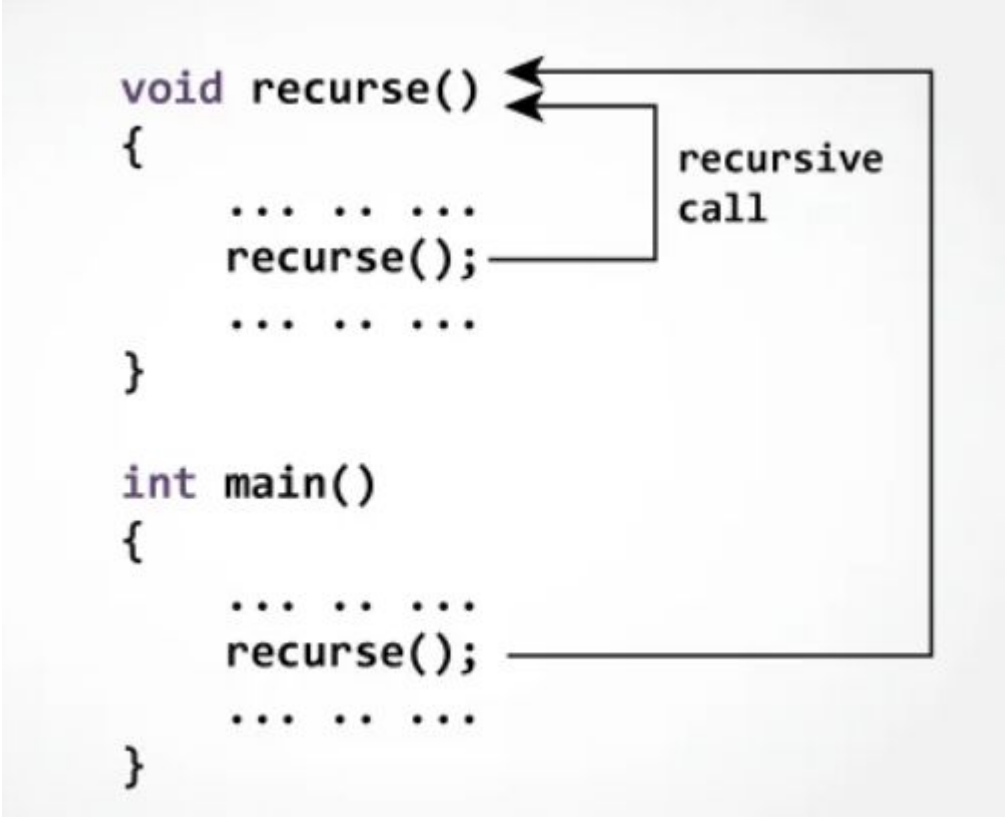
Recursion

➡Recursion is a programming concept where a function calls itself to solve a problem by breaking it down into smaller, simpler versions of the same problem.

➡Recursion is a technique where a function calls itself to solve a problem. It's like having a task and breaking it down into smaller, simpler versions of the same task until you reach a point where the task is so simple that you can solve it directly.

Recursion

- Recursive Call
- Base Case



Factorial using Recursion

5!=5x4x3x2x1

4!=4x3x2x1

3!=3x2x1

2!=2x1

1!=1

Factorial using Recursion

5!=5x4x3x2x1 ➡4!

4!=4x3x2x1 ➡3!

3!=3x2x1 ➡2!

2!=2x1 ➡1!

1!=1

n!=n*(n-1)!

Factorial using Recursion

```
int  factorial(int n){
     if(n==1){
       return 1;
     }
    else
       n*factorial(n-1);
}
```

Problem 1

WAP that will calculate the sum of numbers from 1 to n using recursion.

Sample input	Sample output
5	15
10	55

Problem 2

WAP that will display Fibonacci Series using recursion.

0, 1, 1, 2, 3, 5, 8, 13, 21,.....

Sample input	Sample output
2	0, 1, 1
8	0, 1, 1, 2, 3, 5, 8, 13

Problem 3

WAP that will print the array elements using recursion.

Sample input	Sample output
5 1 2 3 4 5	1 2 3 4 5
3 4 6 8	4 6 8

Problem 4

WAP that will count the digits of a given number using recursion.

Sample input	Sample output
5 1 2 3 4 5	1 2 3 4 5
3 4 6 8	4 6 8

Problem 5

WAP that will get the largest element of an array using recursion

Sample input	Sample output
5 1 2 3 4 5	5
3 4 8 6	8

Problem 6

WAP that will print even or odd numbers in given range using recursion.

Sample input	Sample output
1 10	Even Numbers: 2, 4, 6, 8, 10 Odd Numbers: 1, 3, 5, 7, 9
2 5	Even Numbers: 2, 4 Odd Numbers: 1, 3, 5

Problem 7

WAP that will check whether a given String is Palindrome or not.

Sample input	Sample output
abaaba	Palindrome
abcd	Not Palindrome

Practice more…